

Aaryan Shakya

AI Engineer · Machine Learning & Deep Learning

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PROFESSIONAL SUMMARY

AI Engineer specialized in deep learning architecture, graph neural networks, and custom LLM development. Experienced in training transformer models from scratch, designing multi-agent RAG pipelines, and processing large-scale graph datasets. Proven track record of translating complex ML research into optimized, scalable engineering solutions with a strong focus on clean pipeline design.

CORE TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, HTML/CSS, C, C++

Frameworks & Libraries: PyTorch, TensorFlow, Scikit-learn, LangChain, NumPy, Pandas, Polars

Methodologies: Graph Neural Networks (GNNs), LLMs & Prompt Engineering, Retrieval-Augmented Generation (RAG), Computer Vision (CV), ETL/ELT Pipelines

Tools & Dev Workflow: Git, Docker, n8n (workflow automation), AI coding assistants & Agentic IDEs

KEY TECHNICAL PROJECTS

Multimodal Nepali Cultural Dance Classification Engine | *Machine Learning Engineer*

Engineered a multimodal neural network to classify 20 distinct categories of traditional dance styles. Designed the feature extraction and alignment pipelines to calibrate and synchronize disparate sensory modes into a unified classifier.

LEGIT - Legal Expertise through Generative & Extractive Pre-trained Transformers

Built a 10M-parameter Transformer decoder from scratch on a low-resource ~4M token legal corpus. Achieved 79.4% next-token accuracy, 1.9 perplexity, and a 0.6432 validation loss using mixed-precision training and a warmup-cosine decay schedule.

Large-Scale Data Engineering - OpenAlex Metadata Ingestion | *Technical Lead*

Architected data pipelines to ingest, clean, and map high-dimensional academic metadata from the OpenAlex dataset. Converted unstructured relational records into optimized, ML-ready graph structures for large-scale downstream network analytics.

Enterprise Workflow Automation & Agentic RAG Systems

Designed event-driven automation workflows linking multi-step enterprise data flows across disparate APIs. Engineered an LLM query transformation pipeline using function calling and structured output parsing to inject dynamic data directly into production n8n automation steps.

Audio Emotion Recognition via Adaptive Graph Neural Networks | *GNN*

Reframed speech emotion recognition from traditional 2D spectrograms into a dynamic graph learning execution. Implemented a dyGIN2d (Dynamic Graph Isomorphism Network) architecture to map audio channels as nodes, accurately classifying speech into 7 emotional states.

EmerG-Net - Discovering Growing Communities in Graphs

Developed a predictive framework to identify early-stage, high-potential research clusters before citation saturation. Fused 768-dimensional SciBERT contextual embeddings with TransR structural embeddings to process incremental graph updates via a high-performance FastAPI backend.

RESEARCH & PUBLICATIONS

Deep Reinforcement Learning for Asymmetric Strategy in Baghchal | *Co-author*

- Built and benchmarked four reinforcement learning architectures (MuZero, PPO, DQN, REINFORCE) to play Baghchal, a traditional board game with asymmetric roles between Tiger and Goat pieces.
- Designed a custom PyTorch game environment with role-specific reward functions, tuning separate incentive structures for each side.
- Under MuZero, the Tiger agent reached an 86% win rate and the Goat agent reached 62%, while PPO came out as the most resource-efficient model for deployment.
- Documented emergent multi-agent behaviors in the final writeup, including coordinated pincer movements and tactical sacrifice patterns.

Nepalese Currency Counterfeit Detection System | *Co-author*

- Co-built a computer vision system to verify the authenticity of NPR 500 and NPR 1000 banknotes across 7 security features.
- Compiled and published an open-source dataset of 300+ banknote images, including real-world wear like folding, dirt, and governor-tenure variations.
- Benchmarked statistical methods, SVM, and CNN classifiers against each other, then combined ORB-based region extraction with SSIM scoring for the final pipeline.

EDUCATION

B.Tech. in Artificial Intelligence

Kathmandu University, Dhulikhel, Nepal

Expected Graduation: 2027

LEADERSHIP & MENTORSHIP

Organizing Committee & Mentor | AI Convergence 2026 (KUAIC): Mentored university students through an intensive 3-day data camp covering automated data wrangling with Pandas, statistical visualization with Matplotlib, and introductory ML pipelines.

Hackathon Competitor | KU x Chunjae Education Bootcamp: Completed a rigorous 10-day applied AI sprint and 3-day hackathon, engineering functional machine learning proofs-of-concept under strict time constraints.

LANGUAGES

English: Fluent (Professional Working Proficiency)

Nepali: Native/Bilingual Proficiency

About the job

About Fusemachines

Fusemachines is a leading AI strategy, talent, and education services provider. Founded by Sameer Maskey Ph.D., Adjunct Associate Professor at Columbia University, Fusemachines has a core mission of democratizing AI. With a presence in 4 countries (Nepal, the United States, Canada, and the Dominican Republic) and more than 450 full-time employees, Fusemachines brings global AI expertise to transform companies worldwide. Founded in 2013, Fusemachines is a global provider of enterprise AI products and services, on a mission to democratize AI. Leveraging proprietary AI Studio and AI Engines, the company helps drive the clients' AI Enterprise Transformation, regardless of where they are in their Digital AI journeys. With offices in North America, Asia, and Latin America, Fusemachines provides a suite of enterprise AI offerings and specialty services that allow organizations of any size to implement and scale AI. Fusemachines serves companies in industries such as retail, manufacturing, and government.

Fusemachines continues to actively pursue the mission of democratizing AI for the masses by providing high-quality AI education in underserved communities and helping organizations achieve their full potential with AI.

Role Overview

We're hiring Mid and Senior Data Scientists / ML Engineers to help us build the next generation of intelligent, agentic systems that solve real business problems. At this level, you'll own problems end-to-end: scoping ambiguous business questions with AI Leads, choosing the right modeling approach (classical ML, foundation models, or agentic architectures), shipping to production, and iterating based on real-world signals. Senior candidates will additionally shape technical direction, mentor others, and raise the bar for how the team uses modern AI tooling.

Whether your depth is in NLP, Computer Vision, Multimodal systems, or classical ML, what we care about is your ability to reason from first principles, leverage modern AI tools to multiply your output, and turn ambiguous problems into shipped systems that move the business.

Core Responsibilities

- Own ML projects end-to-end - from problem framing and data exploration through modeling, deployment, monitoring, and iteration.
- Design and ship production ML systems spanning churn and propensity modeling, demand forecasting, anomaly detection, pricing optimization, and recommendation - using the right tool for the job, whether that's a gradient-boosted model, a fine-tuned foundation model, or an agent-based workflow
- Build agentic AI workflows that automate decision-making across the business - including multi-agent systems, tool-using agents, and human-in-the-loop pipelines that combine LLM reasoning with deterministic logic and traditional ML
- Leverage modern AI development tooling (AI coding assistants, agentic IDEs, automated evaluation harnesses, MLOps platforms) as a core part of your engineering practice - not an afterthought
- Apply foundation models and domain-specific ML to extract signals from structured data, text, images, audio, or multimodal inputs depending on the problem at hand
- Design rigorous evaluations for both deterministic models and probabilistic AI systems - offline metrics, online A/B tests, LLM-as-judge frameworks, and regression suites for non-deterministic outputs
- Partner with data and platform engineers to build robust pipelines, feature stores, and serving infrastructure supporting both classical ML and modern AI workloads
- Communicate effectively with technical and non-technical stakeholders - translating model behavior, tradeoffs, and risks into decisions the business can act on
- (Senior) Set technical direction for an area of the ML stack, mentor mid-level and junior engineers, lead design reviews, and influence the team's tooling, evaluation standards, and architectural choices

Technical Requirements

- Bachelor's or Master's degree in Computer Science, Mathematics, Statistics, or a related quantitative field - or equivalent practical experience
- 3+ years of applied data science / ML experience with a track record of shipping models to production. (Senior: 5+ years) including ownership of complex systems and demonstrable technical leadership
- Strong programming proficiency in Python, including modern ML/AI libraries (PyTorch or JAX, scikit-learn, Hugging Face, pandas etc.)
- Solid grounding in statistical modeling, experimentation, and classical ML - regression, tree-based methods (Random Forest, XGBoost, LightGBM), time series forecasting, and anomaly detection
- Hands-on experience with at least one specialty area: NLP and LLMs, Computer Vision, or Multimodal systems. We're domain-agnostic - depth in any of these is valued
- Practical experience working with foundation models and LLM APIs - prompt design, structured outputs, function calling, retrieval-augmented generation (RAG), and fine-tuning where appropriate
- Comfort with modern AI development workflows - using AI coding assistants and agentic IDEs (Claude Code, Cursor, Copilot, or similar) as a daily part of your engineering practice
- Experience with data engineering fundamentals and big-data tooling such as PySpark, Databricks, or equivalent distributed compute frameworks
- Familiarity with MLOps and LLMOps practices - version control for models and prompts, experiment tracking, CI/CD for ML, monitoring, and evaluation pipelines
- (Senior) Demonstrated ability to make architectural decisions, navigate ambiguity, and influence cross-functional partners on technical strategy

Fusemachines is an Equal Opportunities Employer, committed to diversity and inclusion. All qualified applicants will receive consideration for employment without regard to race, color, religion, sex, sexual orientation, gender identity, national origin, disability, or any other characteristic protected by applicable federal, state, or local laws.